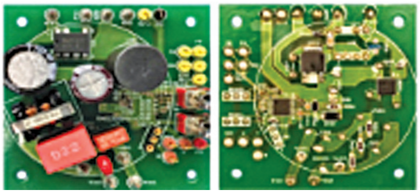
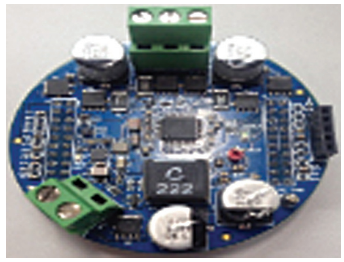
Reference Designs Of MOTOR DRIVES

Title of Reference Design	3-Phase Brushless DC Motor Drive	12V BLDC Motor Drive for Automotive Applications
Image		
Type	3-phase BLDC motor	BLDC
Power	90-264V AC	12V
Wattage	10W	200W
Key Attractions	Reference design for a low-powered BLDC motor drive that can take input from a regular AC home socket (90 to 264 volts) and is capable of running a small 10W motor at 12V DC.	Reference design for a low-powered BLDC motor driver that features an onboard temperature, voltage, and current sense, which allows adding fault-protection and diagnostics features easily to the motor drive.
Highlights	The REFMOT201 is a reference design by ROHM for a 3-phase brushless DC motor drive. The design enables direct input of AC power to a BLDC motor unit by installing an AC-DC converter circuit in the drive circuit for a 3-phase BLDC motor. The design has a 3-phase motor driver plus AC-DC converter (12V output) integrated into a single board. The main advantage of this reference design is its higher efficiency, higher power density, and better torque-to-weight ratio. These features enable the drive to have a smaller footprint and lower volume. The reference design offers precise control over motor speed and torque, making it ideal for applications where accuracy is critical. The low-powered reference design offers higher reliability and low noise and vibration, making it suitable for use in low-powered consumer electronics, such as toys and small fans.	TIDA-00901 is a reference design from the Texas Instruments for a 200W BLDC motor drive. It simplifies the design complexity as it doesn't require any position sensor, such as a quadrature encoder or a hall effect sensor. This design includes an analogue circuit that works with a microcontroller, a C2000 LaunchPad to control the speed of a 3-phase BLDC motor. The design is suitable for harsh automotive environments, such as operation at a wide range of temperatures from –50°C to +150°C. The motor drive can work with a 12V battery and has a wide range of operating voltage from 4.4V to 45V. It is compatible with the existing MotorWare software, thus further simplifying the development process and providing a 3.3V supply for LaunchPad operation.
Application	The reference design is suitable for low-powered consumer electronics and toys.	The reference design is suitable for use in e-bikes, toy cars for kids, etc.
OEM Brand	ROHM	TI
URL	https://www.electronicsforu.com/electronics-projects/industry-grade-projects/reference-design-for-a-three-phase-brushless-dc-motor-drive	—